



TRIBHUVAN UNIVERSITY
FACULTY OF HUMANITIES AND SOCIAL SCIENCES
“Online Eyeglass Shopping System”

A PROJECT PROPOSAL

Submitted to:
Department of Computer Application
Mahendra Morang Adarsh Multiple Campus
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Application*

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1. Introduction

Lensify is a full-featured e-commerce web application designed to provide customers with a convenient and user-friendly platform for browsing, selecting, and purchasing eyewear products, including prescription glasses, sunglasses, and other optical accessories. The system aims to simplify the online shopping experience by offering a well-organized product catalog, secure user authentication, and an efficient order management process.

Inspired by popular online eyewear platforms such as Lenskart, Lensify replicates the essential functionalities of a real-world e-commerce website while demonstrating the practical knowledge and technical skills acquired during the Bachelor of Computer Applications (BCA) programme. The project combines modern frontend technologies to create a responsive and visually appealing user interface with a robust backend developed using PHP and MySQL for database management, user account handling, and order processing.

The platform enables customers to explore products based on categories, compare available options, manage shopping carts, and place orders seamlessly. From an academic perspective, the project serves as a practical implementation of web development concepts, including database design, user authentication, session management, CRUD operations, and e-commerce workflows. Overall, Lensify aims to deliver an efficient online eyewear shopping experience while showcasing the application of modern web technologies in solving real-world business requirements.

2. Problem Statement

The rapid growth of e-commerce has transformed the retail industry, including the eyewear sector. Today, customers prefer online shopping due to its convenience, accessibility, and wider product selection. However, many existing online eyewear platforms still face several limitations that affect the overall user experience. Customers often find it difficult to choose the right eyewear because of the large variety of frames, styles, sizes, and specifications available. Many platforms lack personalized recommendations, advanced filtering options, and efficient order management features. In addition, customers may face challenges such as travel time, limited store hours, and the inconvenience of visiting multiple optical stores before finding suitable eyewear. Existing systems also provide limited support for a smooth and guided purchasing experience, especially for users purchasing prescription glasses. Therefore, there is a

need to develop a modern online eyewear shopping platform that overcomes these limitations by offering a wide range of products, easy accessibility, user-friendly navigation, and an enhanced shopping experience.

Lensify aims to address these challenges by providing a convenient, efficient, and reliable platform for purchasing eyewear products online.

3. Problem Objectives

3.1 Objectives 1

To design and develop a fully functional e-commerce web platform for eyewear products, enabling users to seamlessly browse, filter, and purchase products online through a secure and intuitive shopping experience.

3.2 Objectives 2

To build a secure, role-based admin panel for managing products, orders, and inventory — demonstrating full-stack development proficiency using HTML, CSS, Tailwind CSS, JavaScript, PHP, and MySQL

4. Methodology

4.1 Requirement identification

4.1.1 Literature Review

The concept of e-commerce has evolved significantly over the past two decades, transforming the way consumers interact with retail businesses. What once required a physical visit to a store can now be accomplished within minutes through a web browser or mobile application. This shift has been well-documented across numerous studies, and the eyewear industry is no exception to this trend.

Turban et al. (2015) in their foundational work on electronic commerce describe e-commerce not merely as a sales channel, but as a complete ecosystem involving product discovery, customer engagement, secure transactions, and post-purchase support. Their framework of B2C (Business-to-Consumer) commerce forms the backbone of most online retail platforms — including eyewear stores — where the primary goal is to reduce the friction between a customer's need and the fulfillment of that need. Lensify is designed precisely around this B2C model, where the customer journey from browsing to order placement is kept as seamless as possible.

[1]

In the context of web-based systems, researchers have consistently emphasized the

importance of user interface design and its direct impact on conversion rates and customer satisfaction. Nielsen (1994), whose usability heuristics remain relevant even today, argued that a good system should match the real world, give users control, and maintain consistency throughout the interface. These principles are directly reflected in Lensify's design decisions — categorized product listings, clear navigation, and an intuitive checkout flow are all outcomes of applying these usability standards to an eyewear shopping context. [2]

Database-driven web applications have been extensively studied as a foundation for scalable e-commerce systems. Connolly and Begg (2014) highlight that a well-normalized relational database is critical for maintaining data integrity in systems that handle concurrent users, live inventory, and transactional records. Lensify's database design — built on MySQL with clearly defined entities such as Users, Products, Orders, Cart, and Categories — draws directly from these relational database principles, ensuring that the system can handle real-world data without redundancy or inconsistency. [3]

The role of PHP as a server-side scripting language in e-commerce development has also been widely acknowledged. Welling and Thomson (2016) document PHP's suitability for building dynamic, database-connected web applications, particularly when paired with MySQL. Their work outlines session management, form handling, and CRUD operations — all of which are core technical requirements of the Lensify backend. The choice of PHP 8.2 for this project aligns with their recommendation of using mature, community-supported technologies for academic and small-scale commercial deployments. [4]

Security in web-based systems is another area that has received considerable academic attention. Stuttard and Pinto (2011), in their analysis of web application vulnerabilities, stress that user authentication and session handling are among the most commonly exploited weaknesses in online platforms. This directly informed Lensify's approach to implementing secure login mechanisms, password hashing, and role-based access control — especially for separating customer-facing functionality from the admin panel. [5]

From a Nepali perspective, the growth of digital commerce has been relatively recent but rapid. Studies and reports from the Nepal Rastra Bank and local technology research groups indicate that internet penetration and smartphone usage in Nepal have grown substantially since 2018, creating a viable consumer base for

local e-commerce platforms. However, as observed during the study of existing systems, most locally operating eyewear platforms remain underdeveloped in terms of features, admin infrastructure, and overall user experience. This gap between user expectation and platform capability is what the existing literature collectively points toward as an opportunity for better-designed, locally-relevant systems. [6] Taken together, the literature supports the development of Lensify as both a technically grounded and practically relevant project. The system is not built in isolation — it stands on a well-established body of knowledge in e-commerce design, relational database management, web security, and usability, adapted to meet the specific needs of the Nepali eyewear market.

4.1.2 Requirement Analysis

Taken together, the literature supports the development of Lensify as both a technically grounded and practically relevant project. The system is not built in isolation — it stands on a well-established body of knowledge in e-commerce design, relational database management, web security, and usability, adapted to meet the specific needs of the Nepali eyewear market

a) Functional requirement

Some of the functional requirements of the system are:

- Admin is allowed to add, update and delete products.
- Admin is able to view customer details.
- Admin is able to add different eyeglass brand and design.
- Admin is allowed to add category and subcategory.
- Only admin have access to the user.
- Customer is allowed to view different products.
- Customer is able to order the products.
- Only authentic user must have access to the system

b) Non-functional Requirements

Non-functional requirements are the requirements that specifies how the system performs. a certain function. In other words, a non-functional requirement will describe how a system should behave and what limits there are on its functionality. The non-functional requirements are:

- The maintenance cost of the system must not be very high.
- The interface should be simple so that user can easily navigate through

the website.

- The data must be stored securely that means an unwanted user must not be allowed to exploit the data.
- The response must not be greater than 2 seconds when the active user is less than 2500.
- The data must be backed up so that even after the system gets down it can be retrieved.
- The system must be flexible so that the new requirement can easily be dealt with.

4.2 Feasibility Study

A feasibility study is an assessment of the practicality of the proposed plan or project. According to its workability, impacts on the organization, ability to meet user needs and effective use of the resources. The main tasks done during feasibility study are:

4.2.1 Technical Feasibility

Technical feasibility involves series of questions such as whether the necessary technology and infrastructure needed for the system are available or not, whether the system is compatible or not, how difficult it will be to build with good performance, and whether the system has good security system or not. The system is the fact that it can be developed on any device (windows, mac, linux). The technology or tools used are HTML, CSS, JS, Python and Apache Server. The Google Chrome web browser is used for the testing. So, the system is technically feasible.

4.2.2 Operational Feasibility

The project is operationally feasible as the system only need some of the basic knowledge of computation and internet with a laptop or an android and a browser in order to execute. All these things are common to all and used by everyone in today's world. As the system is user friendly, user can easily access the system. The potential users of this system are stakeholders, such as real estate agents, property developers, or financial institutions owners and their staffs, who are generally familiar with site navigations and management which is the result of advancement of technology in present world, so little or moderate training level would be required. The system will provide a simple interface for the users to operate.

4.2.3 Economic Feasibility

Economic analysis could also be referred to as cost/benefit analysis. Being a web-

based application, this system will use freeware software. It is the most frequently used method for evaluating the effectiveness of a new system. The system which is going to be developed does not require any additional cost and hardware or software as the interface of this system is developed using the resources and technologies which are available freely. So, the system is technically, operationally and economically feasible.

4.3 High Level Design Of System

High Level Design (HLD) is a system design document that includes the description of system architecture, data flows, flowcharts, and data structures, enabling developers and implementers to understand how the system is expected to work with regards to its features and design. Lensify — an Online Eyeglass Shopping System — will be developed using HTML, CSS, JavaScript, PHP, and MySQL. The system will include an administrator panel through which the administrator will have the right to add and delete users, view user information, manage products, and oversee orders. Following are the system design components of Lensify:

1. Login to the system
2. Admin Panel
3. Add user, view history
4. Add, View, Edit and Delete user and their history

4.3.1 Methodology

The methodology of the proposed system involves a systematic approach to develop the System of Online Eyeglass Shopping System. The methodology I would be using for developing this project would be waterfall methodology. The waterfall methodology is a linear and sequential approach to software development, where each phase is completed before moving on to next. It follows a structured flow from requirements gathering to deployment, with limited flexibility for changes once a phase is completed.

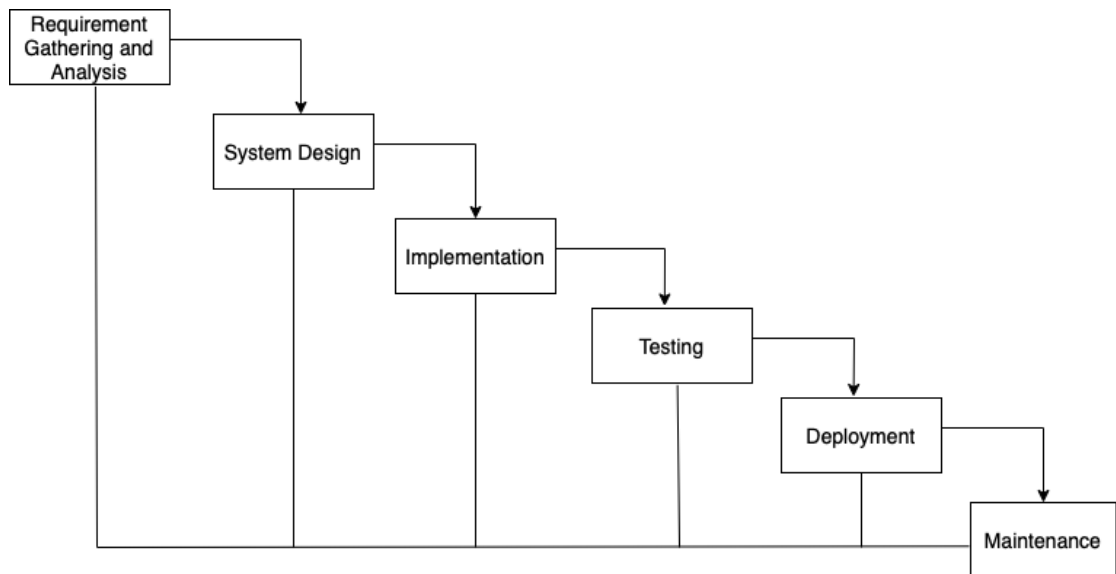


Figure 1 Waterfall Module

4.3.2 System Flowchart

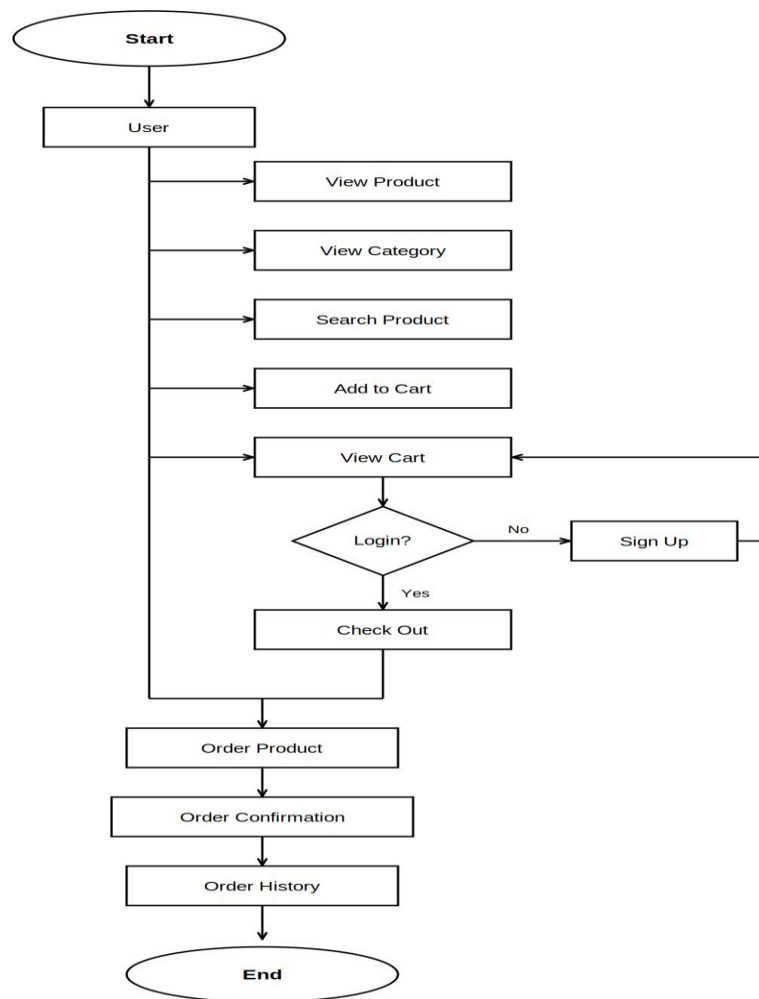


Figure 2 Flowchart Diagram of Online Eyeglass Shopping System

4.3.3 Technology Stack

Table 1 Technology Stack

Layer	Technology	Purpose
Frontend – Structure	HTML5	Page markup and semantic structure
Frontend – Styling	Tailwind CSS, CSS3	Responsive design and utility-first styling
Frontend – Behavior	JavaScript	Dynamic UI interactions and form validation
Backend	PHP 8.2.12	Server-side logic, routing, and session management
Database	MySQL	Relational data storage for users, products, and orders

5. System Module

5.1 User Module

- Handles user registration, login, and logout functionality
- Allows users to edit and update their profile information
- Implements secure session management using PHP
- Manages password storage and authentication system

5.2 Product Module

- Displays product catalogue in categorized format
- Supports product search and filtering options
- Each product has a dedicated detail page
- Shows product images, price, and specifications
- Helps users easily browse eyewear products

5.3 Cart & Order Module

- Allows users to add or remove products from cart
- Manages shopping cart items before checkout
- Supports order placement process
- Stores order details and order history
- Provides basic order tracking functionality

5.4 Admin Module

- Provides a secure admin login system to restrict unauthorized access
- Enables administrators to add, update, and delete product listings
- Manages product categories and inventory details
- Allows viewing and processing of customer orders
- Provides user management functionality (view and manage registered users)
- Displays a dashboard with overall system overview and key statistics
- Ensures controlled management of the entire e-commerce platform

6. Gantt Chart

Table 2 Gantt Chart for Online Eyeglass Shopping System

Weeks →	1	2	3	4	5	6	7	8	9	10	11
Activity ↓											
Planning											
System Design											
Frontend Development											
Backend Development											
Module Integration											
Testing and debugging											
Documentation											

7. Excepted Outcome

The scope of Lensify covers the following functional areas:

- User registration, login, and profile management system
- Product catalogue with categories (prescription glasses, sunglasses, contact lenses)
- Product details page with images, description, and pricing
- Shopping cart and wishlist functionality
- Order placement, order history, and basic order tracking
- Admin dashboard for product management (add, edit, delete)
- Order management and user management for admin
- Responsive design for desktop and mobile devices
- Simulated checkout process (no real payment gateway integration)

8. References

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- [3] T. Connolly and C. Begg, Database Systems: A Practical Approach to Design, Implementation, and Management, Harlow: Pearson Education , 2014.
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